3.3.2 The Product Rule

<u>Motivation:</u> How might we go about finding the first derivative of $f(x) = xe^x$?

One guess might be that we should find the derivative of each factor of *f*. In particular the derivative of *x* is 1 and the derivative of e^x is e^x. So, we might guess that $f'(x) = 1 \cdot e^x$.
However, **THIS TURNS OUT TO BE INCORRECT** (we will find the correct answer below).

To see that there is a problem in our guess above, let's work with a simler function. Consider $g(x) = x^2 = (x)(x)$. We know from the previous section that $g'(x) = 2x$, yet if we try to differentiate both factors we obtain the **INCORRECT** derivative: $g'(x) = 1 \cdot 1 = 1$. This leads us to a very important fact.

Fact: In general $\frac{d}{dx}\left[f(x)g(x)\right] \neq f'(x)g'(x)$

Note: The next theorem will tell us how we should differentiate the product of two functions.

Theorem (The Product Rule): Suppose that *f* and *g* are differentiable functions. Then,

$$\frac{d}{dx}\left[f(x)g(x)\right] = f(x)g'(x) + g(x)f'(x)$$

Note: The product rules tells us to multiply the first factor by the derivative of the second factor, and then add to that the product of the second factor and the derivative of the first factor.

Proof:
Since *f* and *g* are differentiable functions, we know that their first derivatives exist. Using the definition of the derivative we compute:

$$\frac{d}{dx}\left[f(x)g(x)\right] = \lim_{h\to 0}\frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

$$= \lim_{h\to 0}\frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}$$

$$= \lim_{h\to 0}\frac{f(x+h)\left[g(x+h) - g(x)\right] + g(x)\left[f(x+h) - f(x)\right]}{h}$$

$$= \lim_{h\to 0}\frac{f(x+h)\left[g(x+h) - g(x)\right]}{h} + \lim_{h\to 0}\frac{g(x)\left[f(x+h) - f(x)\right]}{h}$$

$$= \lim_{h\to 0} f(x+h) \cdot \lim_{h\to 0}\frac{\left[g(x+h) - g(x)\right]}{h} + g(x)\lim_{h\to 0}\frac{\left[f(x+h) - f(x)\right]}{h} = f(x)g'(x) + g(x)f'(x).$$

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Example 1: Find the first derivative of $f(x) = xe^x$.

Using the *Product Rule* we compute:

$$f'(x) = x\frac{d}{dx}\left(e^x\right) + e^x\frac{d}{dx}(x) = xe^x + e^x \cdot 1 = e^x(x+1)$$

Example 2: Find the first derivative of $f(t) = \sqrt{t}(3-6t)$

Using the *Product Rule* we compute:

$$f'(t) = \sqrt{t}\frac{d}{dt}(3-6t) + (3-6t)\frac{d}{dx}\left(\sqrt{t}\right) = -6\sqrt{t} + (3-6t)\cdot\frac{1}{2}t^{-1/2} = -6\sqrt{t} + \frac{(3-6t)}{2\sqrt{t}}$$

Note: This problem could have also be done by first distributing $\sqrt{t}$ *and then using the Power Rule.*

Example 3: Find the first derivative of $f(y) = \left(y + 2\sqrt{y}\right)e^y$.

Using the *Product Rule* we compute:

$$\frac{df}{dy} = \left(y+2\sqrt{y}\right)\frac{d}{dy}\left(e^y\right) + e^y\frac{d}{dy}\left(y+2\sqrt{y}\right) = \left(y+2\sqrt{y}\right)e^y + e^y\left(1+\frac{1}{\sqrt{y}}\right) = e^y\left(y+2\sqrt{y}+1+\frac{1}{\sqrt{y}}\right).$$

Example 4: Suppose that f, g, and h are differentiable functions. Find a formula for $\frac{d}{dx}\left[f(x)g(x)h(x)\right]$.

To do this, we first view the product of g and h as a single factor and apply the product rule. Then we will apply the product rule again. We have:

$$\begin{aligned}\frac{d}{dx}\left[f(x)g(x)h(x)\right] &= f(x)\frac{d}{dx}\left[g(x)h(x)\right] + g(x)h(x)f'(x)\\ &= f(x)\left[g(x)h'(x) + h(x)g'(x)\right] + g(x)h(x)f'(x)\\ &= f(x)g(x)h'(x) + f(x)h(x)g'(x) + g(x)h(x)f'(x).\end{aligned}$$

Note: The pattern observed here generalizes for more factors.

Example 5: If $f(x)=x^2e^x\sqrt{x}$, then find $f'(x)$.

We use the results observed in the last example to compute this derivative quickly.

$$\begin{aligned}f'(x)&=\frac{d}{dx}\left(x^2\right)e^x\sqrt{x}+x^2\frac{d}{dx}\left(e^x\right)\sqrt{x}+x^2e^x\frac{d}{dx}\left(\sqrt{x}\right)\\&=2xe^x\sqrt{x}+x^2e^x\sqrt{x}+x^2e^x\frac{1}{2\sqrt{x}}\\&=2x^{3/2}e^x+x^{5/2}e^x+\frac{1}{2}x^{3/2}e^x\\&=\frac{5}{2}x^{3/2}e^x+x^{5/2}e^x\end{aligned}$$

Remember that simplifying can make the process of finding a derivative easier.

Notice that: $f(x)=x^2e^x\sqrt{x}=x^{5/2}e^x$ and therefore we have: $f'(x)=\frac{d}{dx}\left(x^{5/2}\right)e^x+x^{5/2}\frac{d}{dx}\left(e^x\right)=\frac{5}{2}x^{3/2}e^x+x^{5/2}e^x$

Example 6: Suppose that *f* and *g* are differentiable functions and that $f(x)=\sqrt{x}\cdot g(x)$, where $g(4)=2$ & $g'(4)=3$. Find $f'(4)$.

We start by finding $f'(x)$ using the *Product Rule*.

$$f'(x)=\frac{d}{dx}\left(\sqrt{x}\right)g(x)+\sqrt{x}\frac{d}{dx}\left(g(x)\right)=\frac{1}{2\sqrt{x}}g(x)+\sqrt{x}\cdot g'(x)$$

Now we will evaluate the derivative at $x=4$, that is we find $f'(4)$.

$$f'(4)=\frac{1}{2\sqrt{(4)}}g(4)+\sqrt{(4)}\cdot g'(4)=\frac{1}{4}\cdot 2+2\cdot 3=\frac{1}{2}+6=6.5\ or\ \frac{13}{2}$$